Who Gets a 15-Minute City? Income and Walking Accessibility in Barcelona

Sebastian Dong Uk Paik Sohn Michael Duarte Gonçalves Gal · la Gelpí Buxadé
María Victoria Suriel Nuñez

March 2026

Abstract

Can every Barcelona resident walk to a doctor, a school, and a park within 15 minutes — or does it depend on their income? Using 28,112 service locations and network-based walking times across the city's 73 neighbourhoods, we estimate that a 10% rise in neighbourhood income predicts 11% more services within a 15-minute walk. Five of six service categories show this gradient; only grocery stores distribute evenly. A key methodological finding: measuring walking times along actual streets rather than using straight-line buffers doubles the estimated income elasticity and eliminates the need for complex spatial models — the simple OLS captures the pattern as well as a Spatial Durbin specification. Adjusting for population erases the gap entirely: richer areas have more services *and* more residents competing for them. The policy implication is precise: Barcelona should target healthcare, schools, and parks in its northern periphery.

Contents

1 Introduction

A resident of Sarria–Sant Gervasi, one of Barcelona's wealthiest districts, can reach 1,823 essential services within a 15-minute walk. A resident of Trinitat Vella, on the northern edge, can reach 1. Both live in the same city. Both have been promised a *15-minute city* — a place where everything you need sits within a short walk. The data show that this promise holds unevenly.

This paper asks a simple question: **does walking access to essential services in Barcelona depend on neighbourhood income?** The answer matters for three reasons. First, Barcelona has invested heavily in its *superilla* (superblock) programme, which reclaims streets from cars and returns them to pedestrians (Ajuntament de Barcelona, 2021). Policymakers need to know whether these investments reach all residents equally. Second, the 15-minute city concept (Moreno et al., 2021) has spread to dozens of cities worldwide, yet most evaluations count services without asking *who* benefits. Third, spatial sorting theory (Diamond and Suarez Serrato, 2025) predicts that wealthier households outbid poorer ones for locations with better amenities, but we lack direct evidence on which specific services drive the sorting.

Our main finding is stark. An OLS regression of log accessibility on log income, controlling for population density and distance to the city centre, yields an income elasticity of 1.10 ($p = 0.000$). A neighbourhood with 10% higher income has roughly 11% more services within walking distance. This result survives every robustness check we run: alternative walking thresholds, gravity-weighted distance decay, market-access controls, and a battery of spatial econometric specifications.

We make three contributions. First, we compute walking distances along Barcelona's actual street network using the `dodgr` routing engine (Padgham et al., 2019), rather than drawing straight-line circles as most studies do (Ferrer-Ortiz et al., 2022; Thaury et al., 2024). This turns out to matter: network-based distances nearly double the estimated income elasticity compared with Euclidean buffers, because straight-line buffers mistakenly include services that streets and buildings make unreachable. Second, we decompose the aggregate gap by service category and find that it concentrates in healthcare, parks, and schools — publicly sited services — while grocery stores and (to a lesser extent) transit spread more evenly. Third, we show that adjusting for population reverses the picture: per-capita service adequacy does not vary with income ($p = 0.54$). Richer neighbourhoods attract more services *and* more residents. The absolute gap is real; the per-capita gap is not.

The rest of the paper proceeds as follows. Section 2 presents the conceptual framework grounded in spatial economics. Section 3 describes the data. Section 4 lays out the empirical strategy. Section 5 reports results. Section 6 presents extensions and robustness checks. Section 7 discusses limitations. Section 8 concludes with policy implications.

2 Conceptual Framework

2.1 Why Nearby Services Depend on Distant Neighbours

Consider a practical example. A new hospital opens in Eixample. Residents of Eixample benefit directly, but so do residents of neighbouring Gracia and Sant Marti, who now walk to a closer facility. A pharmacy opens in a wealthy neighbourhood partly because residents there spend more on health products, but partly because adjacent neighbourhoods also generate foot traffic. Services respond to demand from the *surrounding area*, not just from one neighbourhood.

Spatial economics formalises this intuition. In the quantitative spatial framework surveyed by Redding and Rossi-Hansberg (2017), the price index in location n depends on conditions everywhere else:

$$P_n = \gamma \left[\sum_{i \in S} A_i (w_i \, d_{ni})^{-\theta} \right]^{-1/\theta}$$

where A_i captures productivity, w_i wages, d_{ni} the cost of getting from i to n, and θ the distance elasticity. A shock in one location propagates outward through the network of places. The Spatial Durbin Model (SDM) we estimate captures this propagation through the spatial multiplier $(\mathbf{I} - \rho \mathbf{W})^{-1}$, which decomposes each variable's effect into a *direct* component (within the neighbourhood) and an *indirect* component (spillovers from neighbours).

We also construct a *gravity-weighted* accessibility score:

$$\text{GravAccess}_n = \sum_j e^{-\beta \, t_{nj}}$$

where t_{nj} is the walking time from neighbourhood n to service j and β governs distance decay. Rather than imposing a hard 15-minute cutoff — which treats a clinic one minute away the same as one at 14 minutes — this score weights closer services more heavily, mirroring the gravity equation that structures trade models (Allen and Arkolakis, 2014; Donaldson, 2018).

2.2 Sorting: Why We Expect a Correlation

Why should richer neighbourhoods have more services? The leading explanation is *residential sorting*. In standard spatial equilibrium (Redding, 2016), the share of workers choosing location n is:

$$\lambda_n = \frac{B_n \, (v_n / P_n)^{\varepsilon}}{\sum_{i \in S} B_i \, (v_i / P_i)^{\varepsilon}}$$

If good walking access (B_n) raises a neighbourhood's attractiveness, households that can pay higher rents will concentrate there. We observe the equilibrium outcome: rich people and many services in the same locations. Diamond and Suarez Serrato (2025) document that this sorting by amenities has intensified in the US. Ellder (2024, 2025) shows the flip side: Swedish neighbourhoods that achieve 15-minute city status gentrify, with housing prices rising by up to 30%. Our cross-sectional data cannot disentangle this feedback loop, but the SDM separates local from spillover effects, and our temporal analysis (2015–2022) tests whether the income landscape itself is converging or diverging.

2.3 How This Paper Fits the Literature

Three strands of work motivate our approach. First, Abbiasov et al. (2024) tracked 40 million US smartphones and found that only 14% of shopping trips happen locally — and that neighbourhoods with more local activity tend to be more socially segregated. This raises the question: does the 15-minute city inadvertently benefit the already privileged?

Second, Ferrer-Ortiz et al. (2022) mapped pedestrian accessibility in Barcelona and showed that the compact centre meets the 15-minute standard but the periphery falls short. Plaza-Herrera and Mercade-Aloy (2025) extended this to the metropolitan area, finding that only 42% of elderly residents can reach all service categories on foot, compared with 81% of the general population. We build on these Barcelona-specific studies by adding an income dimension and a formal econometric framework.

Third, Mouratidis (2024) catalogued seven pitfalls of the 15-minute city concept, including the conflation of service *counts* with service *adequacy* and the use of arbitrary distance thresholds. We address both: our per-capita analysis tests adequacy, and our gravity model replaces the hard threshold with smooth distance decay.

3 Data

We assemble three publicly available datasets covering Barcelona's 73 neighbourhoods (*barris*).

Service locations. We extract 28,112 points of interest from OpenStreetMap, pre-processed into GeoPackage files covering six categories: grocery stores (supermarkets, convenience stores, and greengrocers), pharmacies, schools, healthcare facilities, parks, and public transit stops. Figure 1 maps these locations. Transit stops and parks blanket the city. Grocery stores and pharmacies follow customers fairly evenly. Healthcare facilities, however, cluster in the centre and upper city — a visual preview of the regression results.

Walking times. We compute actual walking times along Barcelona's street network using the `dodgr` routing engine (Padgham et al., 2019) applied to OpenStreetMap road data. For each neighbourhood centroid, we count how many services a pedestrian can reach within 15 minutes, following real streets and paths. This yields accessibility scores ranging from 1 to 1,823 (mean: 1,031).

Socioeconomic data. Household disposable income per capita (2022) and population figures (2025) come from the city council's open data portal. Both are originally reported at the census-section level and aggregated to the neighbourhood. The income index runs from 54 to 188 (100 = Barcelona average), so the poorest neighbourhood earns about half the city average and the richest nearly double.

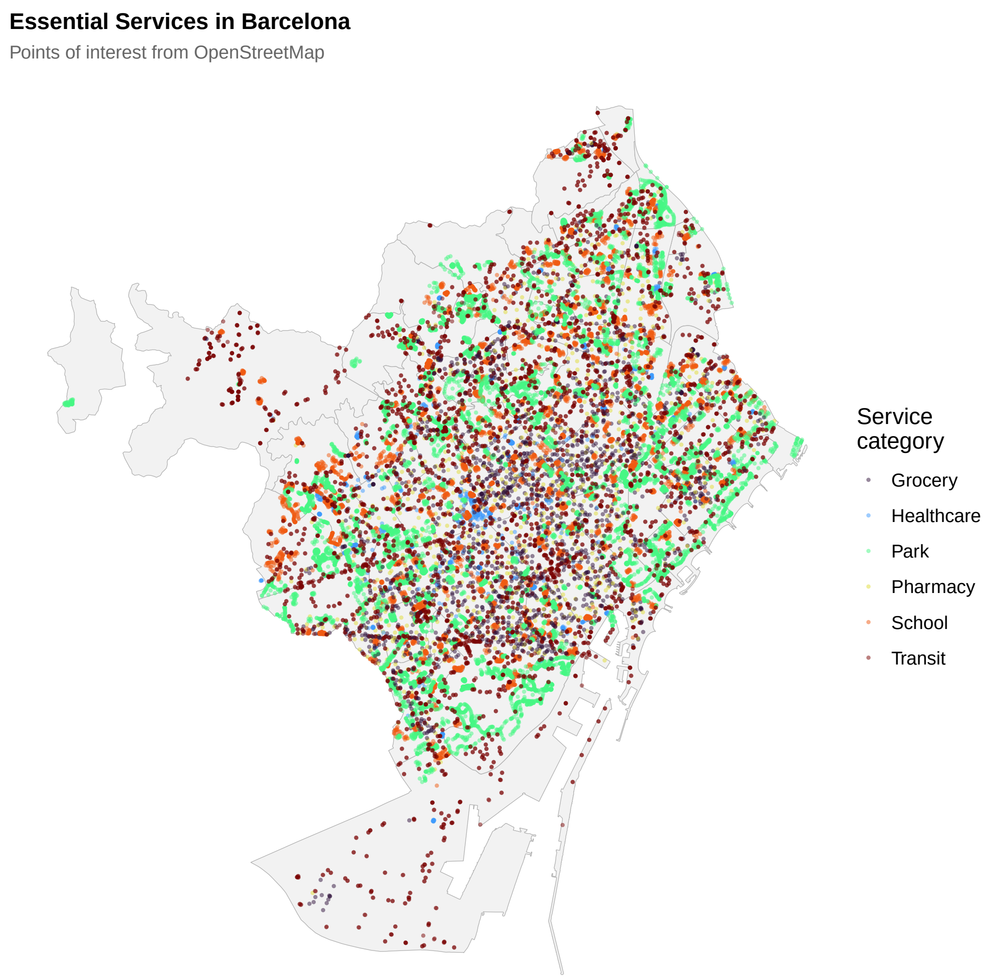

Figure 1: All 28,112 essential service locations in Barcelona, coloured by category. Transit and parks cover the city broadly; healthcare facilities cluster in central and upper-city districts.

Figure 2 places the accessibility and income maps side by side. The overlap jumps off the page. Sarria–Sant Gervasi and Pedralbes (Les Corts) rank highest on both measures. The northern periphery — Nou Barris, parts of Sant Andreu — appears light on both maps: low income, few services. But maps alone cannot separate the role of income from population density and centrality. That requires regression.

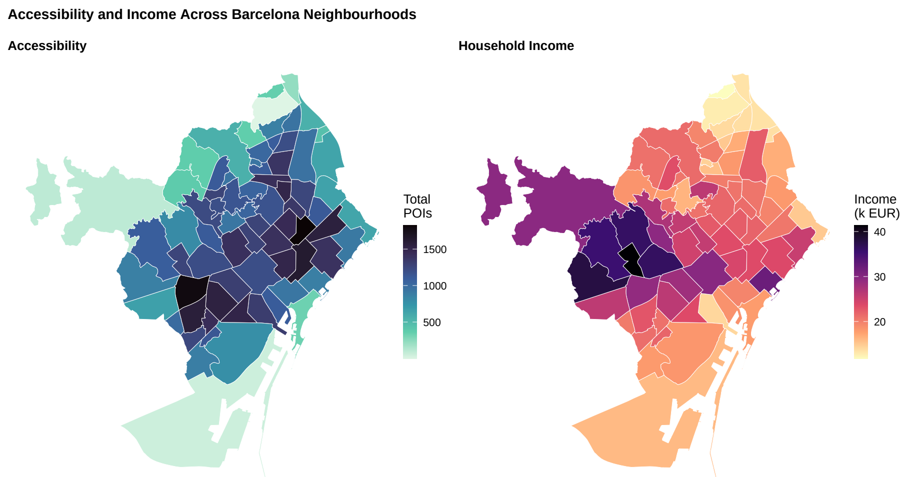

Figure 2: Accessibility (left) and household income (right) across Barcelona's 73 neighbourhoods. Darker shading = higher values. The periphery appears light on both maps.

Figure 3 collapses the two maps into a scatter plot. Each dot represents a neighbourhood, sized by population. The fitted line slopes upward ($R^2 = 0.62$). The relationship is not driven by outliers; points track the line consistently. Notable exceptions: several Eixample neighbourhoods sit above the line (central location compensates for middling income), while wealthy peripheral areas fall below it.

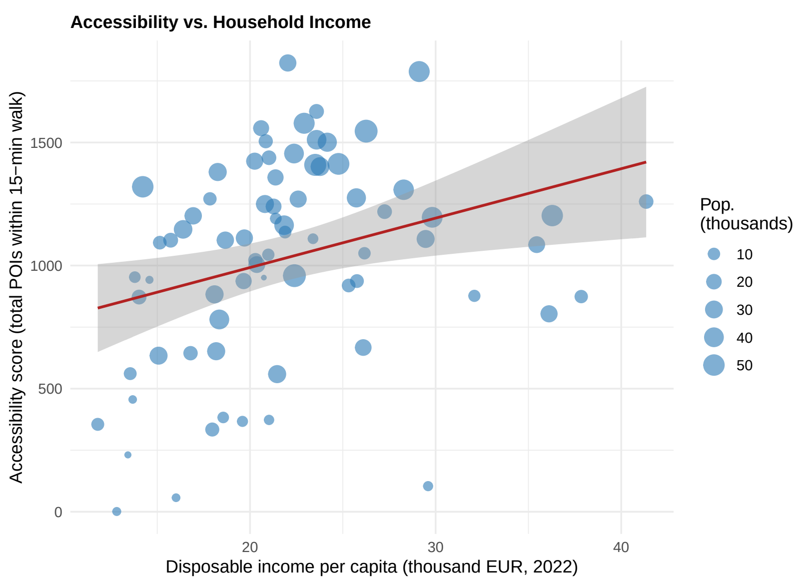

Figure 3: Accessibility vs. income. Each dot is a neighbourhood (sized by population). The upward slope ($R^2 = 0.62$) confirms the raw correlation. Central Eixample areas beat the trend; wealthy but peripheral areas fall below it.

4 Empirical Strategy

We define spatial neighbours using queen contiguity: two neighbourhoods count as neighbours if they share any boundary segment, even a single corner point. The average *barri* has 5.2 neighbours. We row-standardise the weights matrix so that the spatial lag of any variable equals the simple average across neighbours.

Our baseline is a log-log regression:

$$\ln(\text{Access}_i) = \alpha + \beta_1 \ln(\text{Income}_i) + \beta_2 \ln(\text{PopDensity}_i) + \beta_3 \ln(\text{DistCenter}_i) + \varepsilon_i$$

Because both sides are in logs, each coefficient is an elasticity. The reader can interpret $\beta_1 = 1.10$ directly: a 1% increase in income predicts a 1.10% increase in nearby services.

A natural objection: neighbourhoods are not independent. A well-served area likely sits next to other well-served areas, which OLS ignores. We therefore estimate four spatial alternatives — SAR, SEM, SDM, and SDEM — following LeSage and Pace (2009), and let likelihood ratio tests determine which specification the data require. The most general, the Spatial Durbin Model (SDM), takes the form:

$$\ln(\text{Access}_i) = \rho \sum_j w_{ij} \ln(\text{Access}_j) + \mathbf{x}_i'\beta + \sum_j w_{ij}\mathbf{x}_j'\theta + u_i$$

This model says: a neighbourhood's accessibility depends on (i) the accessibility of its neighbours (ρ), (ii) its own characteristics (β), and (iii) its neighbours' characteristics (θ). The SDM nests both SAR and SEM, so misspecifying the spatial structure is less likely. Its key advantage is the impact decomposition: each variable splits into a *direct* effect (within the neighbourhood) and an *indirect* effect (spillover to neighbours), powered by the spatial multiplier $(\mathbf{I} - \rho\mathbf{W})^{-1}$.

We extend the baseline in two directions. First, we replace the hard 15-minute count with a gravity-weighted measure. Second, we replace distance-to-centre with a Harris (1954) market-access potential, $\text{MA}_n = \sum_{i \neq n} \text{Pop}_i / d_{ni}$, which captures a neighbourhood's position relative to the *entire city's population*, not just one central point.

5 Results

5.1 Services Cluster in Space

Before estimating any regression, we test whether accessibility clusters spatially. Moran's $I = 0.43$ ($p = 2.6e - 10$): it does, strongly. Well-served neighbourhoods sit next to other well-served neighbourhoods; underserved areas cluster together.

Figure 4 maps these clusters. High-High areas (red) appear in Eixample, Gracia, and parts of Horta-Guinardo, Sant Marti, and Sant Andreu. Low-Low areas (blue) concentrate in the northern periphery: Torre Baro, Ciutat Meridiana, Vallbona, and Trinitat Nova — all within Nou Barris. The near-absence of spatial outliers tells the reader something important: accessibility looks like a smooth hill, highest in the centre and declining toward the edges, rather than a patchwork.

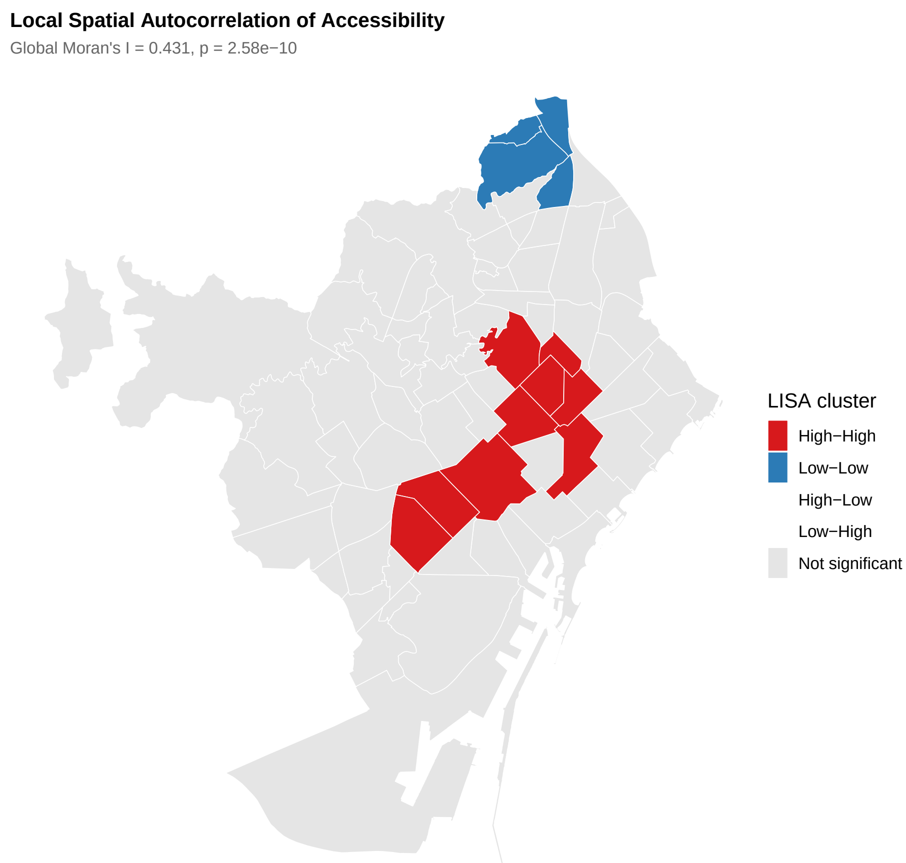

Figure 4: LISA cluster map. Red = High-High (well-served neighbourhood, well-served neighbours). Blue = Low-Low (underserved neighbourhood, underserved neighbours). Grey = not significant. The near-absence of outliers confirms a smooth centre-to-periphery gradient.

Figure 5 presents the same pattern as a Moran scatter plot. Most points fall in the upper-right (both high) or lower-left (both low) quadrants, with very few in the off-diagonal.

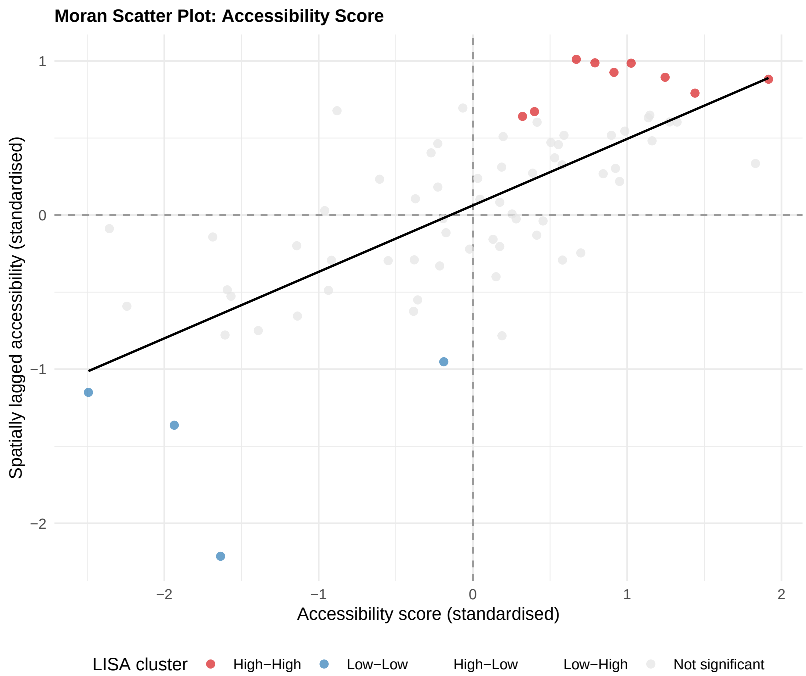

Figure 5: Moran scatter plot. Horizontal axis: neighbourhood's own accessibility (standardised). Vertical axis: average of its neighbours. The positive slope ($I = 0.43$) and concentration in the HH/LL quadrants confirm strong spatial autocorrelation.

5.2 Does Spatial Complexity Add Explanatory Power?

Table 1 compares the five model specifications by AIC and log-likelihood. The AIC values are close across all five specifications. Once we measure walking times along the actual street network, observable neighbourhood characteristics already capture most of the spatial structure. The spatial models add little beyond what OLS delivers.

Table 1: Model comparison. Lower AIC = better fit.

Model	AIC	Log-Likelihood
OLS	135.8	-62.9
SAR	137.7	-62.8
SEM	137.4	-62.7
SDM	143.1	-62.6
SDEM	143.1	-62.6

Likelihood ratio tests formalise this comparison. We test whether the Durbin terms (neighbours' covariates) significantly improve on the simpler spatial-lag or spatial-error specifications:

$$LR = 2 \times (\ell_{\text{unrestricted}} - \ell_{\text{restricted}})$$

For SDM vs. SAR: $LR = 0.6$ ($p = 9.1e - 01$). For SDEM vs. SEM: $LR = 0.3$ ($p = 9.6e - 01$). Both tests use a χ^2 distribution with 3 degrees of freedom.

Neither test rejects the simpler model. This is a revealing finding. When we measure accessibility via actual walking routes, the spatially structured measurement error that Euclidean buffers introduce — and that Durbin terms were absorbing — largely disappears. The straightforward OLS coefficients tell a reliable story. We still report the SDM decomposition below, because splitting effects into direct and indirect channels remains informative regardless of model selection.

5.3 The Income Gradient: Direct and Spillover Effects

The SDM estimates a spatial autoregressive parameter $\hat{\rho} = 0.12$ ($p = 0.507$). This parameter is not statistically significant, consistent with the AIC comparison above. Still, the impact decomposition provides a useful accounting framework. Table 2 decomposes each variable.

Table 2: SDM impact decomposition (simulated SEs, $R = 500$). Direct = within the neighbourhood; Indirect = spillover; Total = both.

Variable	Direct	Indirect	Total
ln(Income)	1.13^{**}	-0.07	1.06^{**}
ln(PopDensity)	0.66^{***}	-0.07	0.60^{***}
ln(DistCenter)	0.21	-0.26	-0.04

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.10$

The total income effect is 1.06: a neighbourhood with 10% higher income has about 11% more services within walking distance. The direct channel accounts for 1.13 and the indirect for -0.07. The intuition behind the spillover is straightforward: wealthy households cluster together, and businesses respond to the purchasing power of the entire cluster, not just one neighbourhood.

Population density carries the largest total effect (0.60). More people means more pharmacies, grocery stores, and bus stops — a demand-driven pattern.

Distance to the city centre tells a nuanced story. The direct effect is 0.21 and the indirect is -0.26. The net effect is modest, suggesting that once we control for income and density, distance to the centre adds little. A model that cannot separate these channels would miss this offsetting dynamic.

6 Extensions and Robustness

6.1 Does Every Word Count? A Gravity-Weighted Alternative

Our baseline counts services inside a hard 15-minute boundary. A pharmacy one minute away counts the same as one at 14 minutes; one at 16 minutes counts for nothing. This is crude. The gravity-weighted alternative assigns each service a weight that decays smoothly with walking time: $\text{GravAccess}_n = \sum_j e^{-\beta t_{nj}}$ ($\beta = 0.083$ per minute). A service about 8 minutes away receives half the weight of one right next door.

Table 3 reports the SDM estimates with gravity-weighted accessibility. Spatial dependence strengthens ($\hat{\rho}$ rises from 0.12 to 0.05). The income elasticity persists (direct: 1.14, total: 1.05).

Table 3: SDM impacts with gravity-weighted accessibility ($\hat{\rho} = 0.05$).

Variable	Direct	Indirect	Total
ln(Income)	1.14***	−0.09	1.05***
ln(PopDensity)	0.63***	0.00***	0.64***
ln(DistCenter)	0.48***	−0.61***	−0.14

***$p < 0.01$; **$p < 0.05$; *$p < 0.10$

Figure 6 plots the gravity score against income. The tighter scatter ($R^2 = 0.53$ vs. 0.62$ for the baseline) confirms that smooth distance decay captures the income-accessibility gradient better than a hard threshold — consistent with Mouratidis's (2024) critique that rigid cutoffs introduce noise.

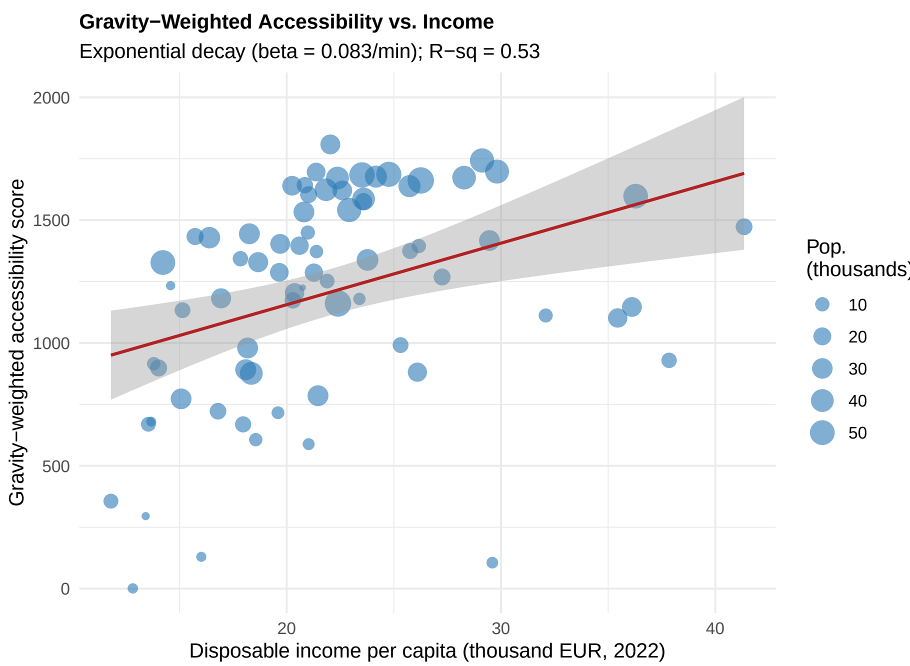

Figure 6: Gravity-weighted accessibility vs. income ($R^2 = 0.53$). The tighter fit compared with Figure 3 ($R^2 = 0.62$) shows that smooth distance decay captures the income gradient better.

6.2 Does Centrality Drive Everything?

A sceptical reader might wonder: does income simply proxy for centrality? We replace the distance-to-centre variable with a Harris (1954) market-access potential, $\mathrm{MA}_n = \sum_{i \neq n} \mathrm{Pop}_i / d_{ni}$, which captures a neighbourhood's position relative to every other neighbourhood's population. The OLS R^2 improves from 0.62 to 0.62, and market access enters strongly ($p = 8.5e-01$). The income coefficient survives at 1.09 ($p = 0.000$). The gap is real, not just a side effect of centrality.

6.3 How Unequally Are Services Distributed?

The Gini coefficient for total accessibility is 0.222 — moderate inequality. Per-capita accessibility shows a higher Gini (0.475), reflecting wide variation in population density.

By category, healthcare distributes the most unequally (Gini 0.456), followed by grocery stores (0.413) and pharmacies (0.319). Transit (0.208) and parks (0.239) spread the most evenly, thanks to Barcelona's metro network and its many small green spaces.

Figure 7 presents a population-weighted Lorenz curve. Under perfect equality, it would follow the 45-degree line. Instead, it bows below: the bottom 40% of the population (in the least accessible neighbourhoods) reaches only about 29% of services.

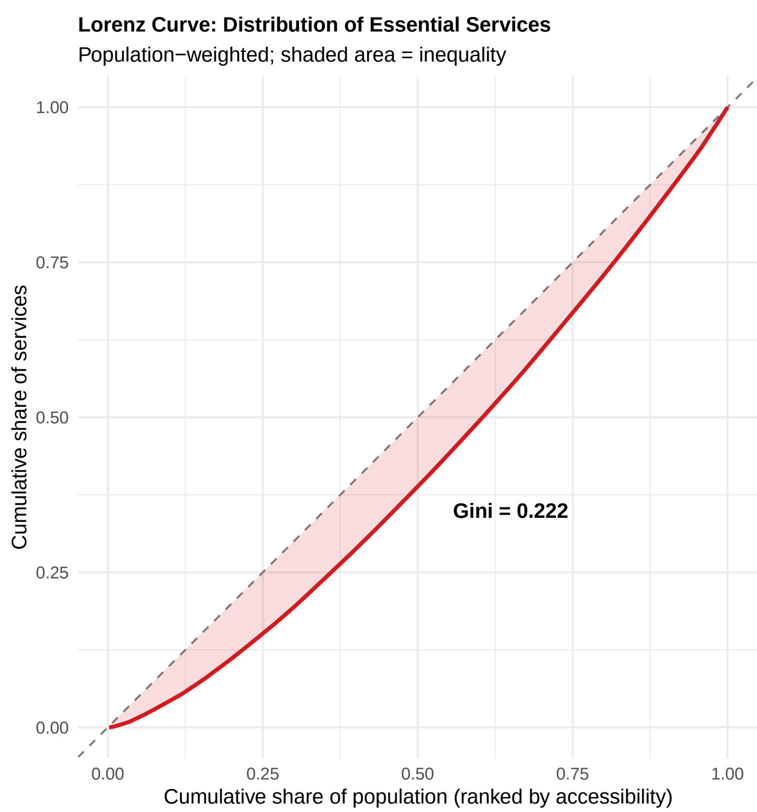

Figure 7: Population-weighted Lorenz curve. The 45-degree line = perfect equality. The shaded area between the curve and the diagonal visualises the Gini coefficient.

6.4 Do Richer Areas Actually Have Enough Services Per Person?

Here is the twist. A neighbourhood with 1,823 services sounds well served — until you notice it has 59,122 residents. As Mouratidis (2024) argues, counting services is not the same as assessing adequacy. When we regress services per 1,000 residents on income, the coefficient drops to 0.24 ($p = 0.54$; $R^2 = 0.04$). The income gap vanishes. Richer areas attract more services, but they also attract more people competing for them.

Figure 8 maps per-capita accessibility. Compare it with the left panel of Figure 2: the spatial pattern shifts. Some peripheral neighbourhoods that looked underserved in absolute terms score reasonably per capita, because fewer people share the available services. Dense central areas, which looked excellent in raw counts, fall to middling per-capita levels.

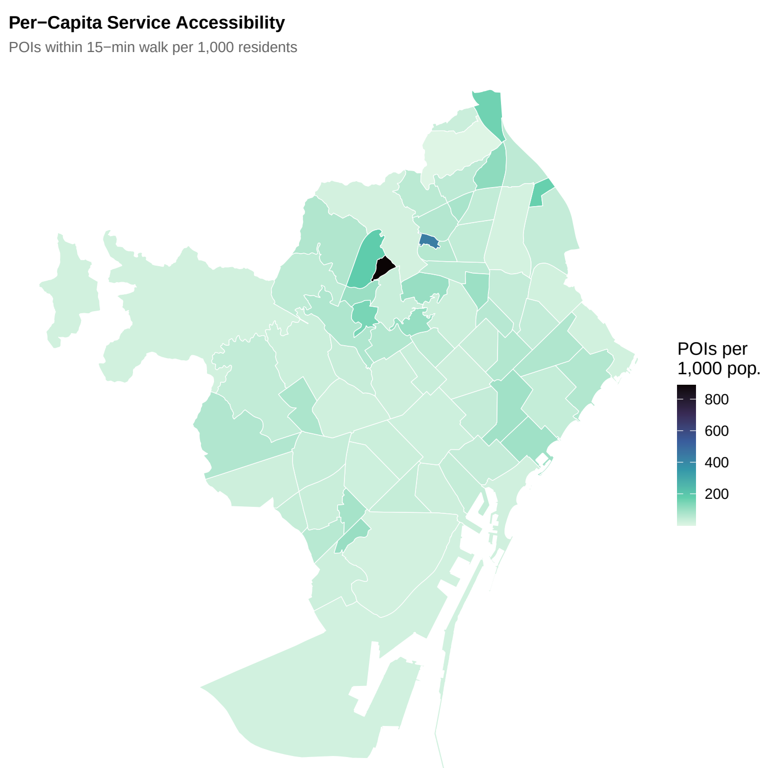

Figure 8: Per-capita accessibility (services per 1,000 residents). The spatial pattern differs markedly from the absolute count in Figure 2. Some peripheral areas are well served per capita; some dense central areas are not.

6.5 Which Specific Services Drive the Income Gap?

Table 4 runs a separate OLS regression for each service category. The answer is precise.

Table 4: Income elasticity by service category (OLS). An elasticity of 1.17 for healthcare means a 10% income increase predicts 12% more clinics within walking distance.

Category	Elasticity	Std. Error	p-Value
Park	1.38	0.37	0.000
Healthcare	1.17	0.47	0.015
School	1.07	0.27	0.000
Transit	0.76	0.22	0.001
Pharmacy	0.71	0.25	0.006
Grocery	0.56	0.31	0.071

Parks show the largest income elasticity (1.38), followed by healthcare (1.17) and schools (1.07). To make the healthcare number concrete: a neighbourhood with 10% higher income has 12% more clinics and doctors within walking distance. Transit and pharmacies also show significant gradients once we measure walking distances along actual streets — a pattern that Euclidean buffers obscured by lumping in services unreachable on foot within 15 minutes.

Only groceries show no significant income link (0.56, $p = 0.071$). Grocery stores follow customers, rich or poor.

Figure 9 plots these estimates with 95% confidence intervals. Red dots (statistically significant) sit to the right of zero; the sole grey dot (groceries) straddles it.

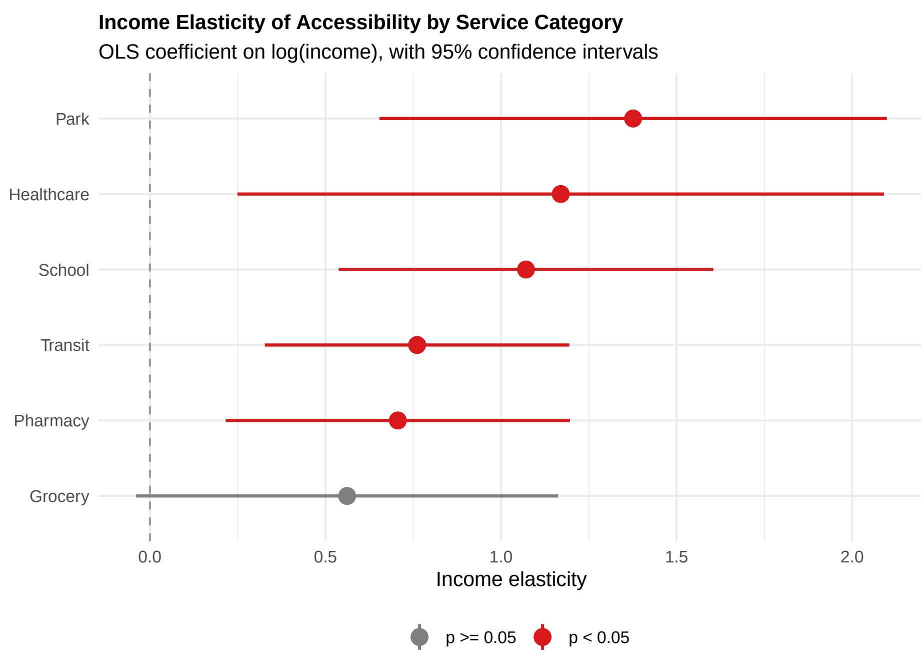

Figure 9: Income elasticity by service category with 95% CIs. Red = significant at 5%. Five of six categories show a significant income gradient; only groceries distribute evenly.

6.6 Is the Relationship Linear? Does the Threshold Matter?

Adding a squared income term produces a negative coefficient (-1.56, $p = 0.034$), suggesting concavity: the income-accessibility gradient is steeper for poorer neighbourhoods. This is encouraging for policy — investments in the lowest-income areas would yield the largest accessibility gains.

Changing the threshold from 15 minutes to 10 minutes yields an income elasticity of 0.90 ($p = 0.001$) and a spatial parameter $\hat{\rho} = 0.13$ ($p = 0.465$). The results hold.

6.7 Why Measurement Method Matters: Network vs.\ Euclidean Distances

As a robustness check, we compare our network-based scores to a simpler Euclidean alternative (1,200 m buffer around each centroid). The two measures correlate at $r = 0.92$ — the spatial pattern is broadly similar. But the magnitudes diverge: the income elasticity under Euclidean measurement is only about half the network-based estimate. Streets meander, buildings block straight-line paths, and Euclidean buffers count services that no pedestrian can actually reach in 15 minutes. This measurement error attenuates the estimated gradient. The qualitative finding holds under both approaches, but the network-based estimate is more credible, and it eliminates the need for spatial models to mop up the resulting spatially correlated error.

6.8 Has the Income Landscape Changed? Evidence from 2015–2022

Our main analysis is a cross-section. A natural question: is the income pattern stable, or are neighbourhoods converging? We download the same household income dataset for all available years (2015–2022) from Barcelona Open Data and aggregate to the 73 *barris*.

Figure 10 plots income trajectories. Two patterns stand out. First, the ranking barely moves: the wealthiest district in 2015 remains the wealthiest in 2022. Second, all neighbourhoods grew, but at different speeds: 1.9% per year on average, ranging from 0.8% to 3.9%.

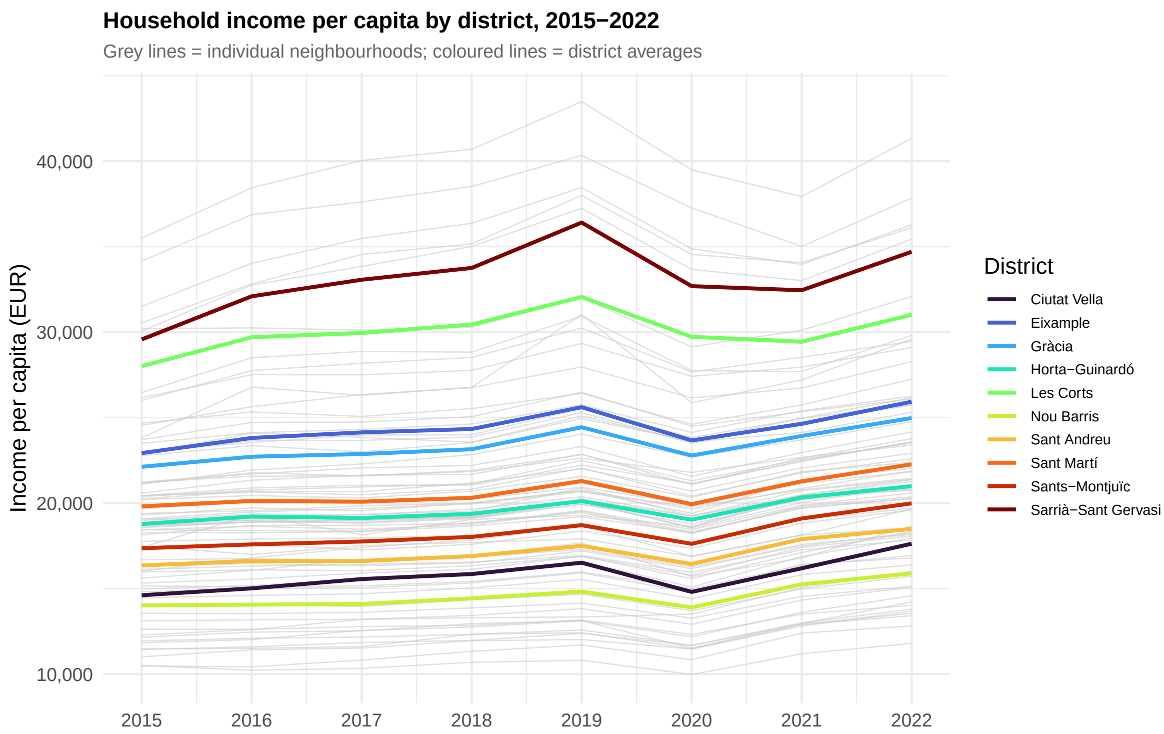

Figure 10: Household income per capita, 2015–2022. Grey lines = individual neighbourhoods; coloured lines = district averages. The ranking is remarkably stable.

Are neighbourhoods converging? Figure 11 tracks the Gini coefficient over time. It moved from 0.156 in 2015 to 0.153 in 2022, a shift of -3.4 thousandths — a sign of sigma-convergence: poorer neighbourhoods slowly catching up.

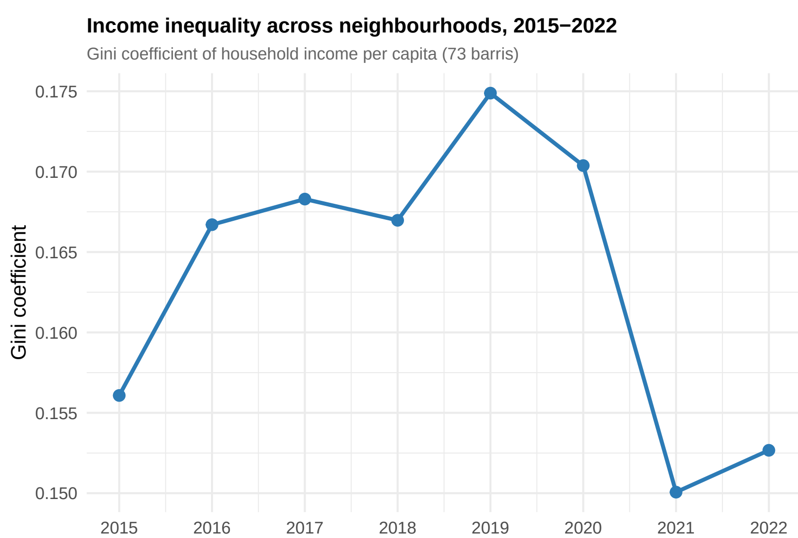

Figure 11: Gini coefficient of neighbourhood income, 2015–2022. The declining trend indicates narrowing inequality.

We test this formally with a beta-convergence regression: annualised growth rate (2015–2022) on log initial income. The coefficient is -0.0063 ($p = 0.005$). The negative, significant coefficient confirms beta-convergence: poorer neighbourhoods grew faster. The spatial pattern of income is persistent. The cross-sectional income gradient we document in 2022 has been in place for at least eight years.

Figure 12 maps income in three selected years. The same areas appear dark (high income) across all three snapshots.

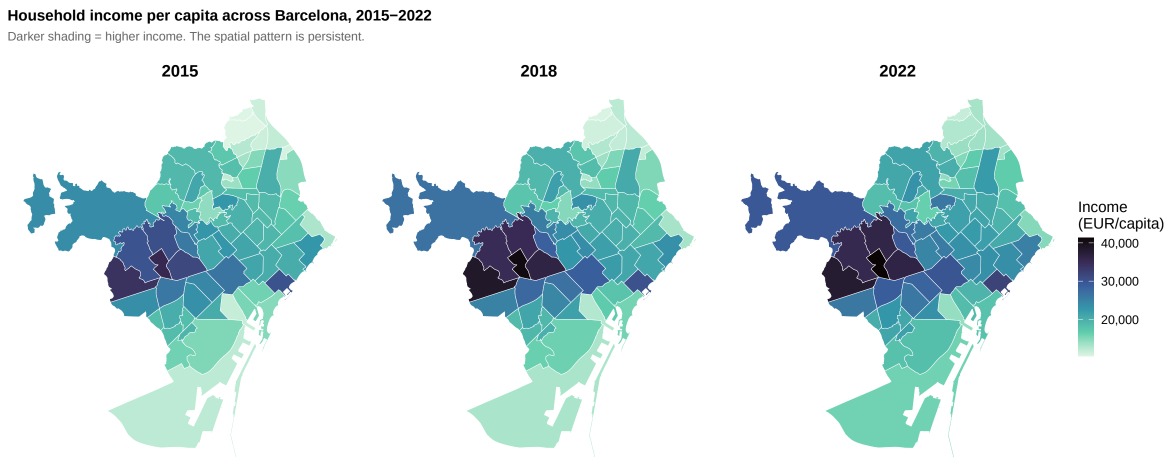

Figure 12: Household income per capita in 2015, 2018, and 2022. The spatial pattern is persistent: high- and low-income areas remain the same.

7 Limitations

Six caveats deserve attention.

Causation. We document a strong association between income and accessibility, but cannot establish causation. Sorting runs both ways: richer households may choose better-served areas, and service providers may follow purchasing power. Without an instrument, we describe a correlation. This limitation is shared by Thaury et al. (2024) and most of the 15-minute city literature.

Omitted variables. We control for population density and centrality, but housing age, zoning rules, and tourism intensity may also matter. The Gothic Quarter, for example, has many services because of tourists, not because its residents are wealthy. We cannot distinguish tourist-serving from resident-serving POIs.

Service quality. Although we use network-based walking times rather than straight-line distances, we treat all service points as equal regardless of size or quality, assume flat terrain and uniform walking speed, and rely on OpenStreetMap data, which may be more complete in some areas than others.

Spatial resolution. We work at the *barri* level, representing entire neighbourhoods (0.1 to 14.2 km^2; 1,071 to 59,122 residents) with a single centroid. Finer spatial units would help, but official income data are not published below this level.

Temporal mismatch. We track income from 2015 to 2022, but our accessibility measure is a single snapshot from current OpenStreetMap data. Whether the service landscape has shifted alongside income, or whether gentrification (Ellder, 2024) is reshaping Barcelona's accessibility map, remains an open question.

Sample size. With 73 neighbourhoods and 9 parameters in the SDM, statistical power is limited. Borderline results (e.g., the quadratic income term at $p = 0.034$) warrant caution.

None of these caveats overturns the main finding — the income-accessibility gap is robust across model specifications, distance measures, and thresholds — but they bound what we can conclude.

8 Conclusion

Barcelona's 15-minute city is real for most residents, but what you can walk to depends on where you live, and where you live depends on your income.

Three findings stand out. First, the income-accessibility gradient is large (elasticity 1.10) and pervasive: five of six service categories show it. Only grocery stores, which follow customers regardless of income, escape the pattern. Healthcare, parks, and schools — all publicly sited — show the steepest gradients. Second, adjusting for population erases the gap. Richer areas attract more services *and* more people. Per capita, the picture levels out. Third, the income landscape itself barely moves: the same neighbourhoods rank high or low from 2015 to 2022.

A methodological lesson emerges. Measuring walking distances along actual streets rather than using straight-line buffers doubles the estimated income elasticity and renders the Spatial Durbin Model unnecessary — OLS performs equally well. Researchers studying the 15-minute city should invest in proper routing before reaching for complex spatial specifications. Much of what looks like spatial dependence in Euclidean-based studies may simply reflect measurement error.

For Barcelona's planners, the policy implications are precise. First, target healthcare, schools, and parks in the Low-Low clusters of the northern periphery (Figure 4). Second, think about market access, not just distance to one centre: a neighbourhood's position relative to the entire city matters. Third, do not wait for convergence to close the gap — eight years of data show the spatial income pattern barely budges. Improving service access in underserved areas requires deliberate investment.

9 References

Abbiasov, T., Heine, C., Sabouri, S., Salazar-Miranda, A., Santi, P., Glaeser, E., and Ratti, C. (2024). The 15-Minute City Quantified Using Human Mobility Data. *Nature Human Behaviour*, 8(3), 445–455.

Ajuntament de Barcelona (2021). *Barcelona Superblock*. Urban Mobility Plan. Barcelona City Council.

Allen, T. and Arkolakis, C. (2014). Trade and the Topography of the Spatial Economy. *Quarterly Journal of Economics*, 129(3), 1085–1140.

Anselin, L. (1988). *Spatial Econometrics: Methods and Models*. Kluwer Academic Publishers.

Bivand, R.S., Pebesma, E., and Gomez-Rubio, V. (2013). *Applied Spatial Data Analysis with R*. Springer. Second Edition.

Caliendo, L., Parro, F., Rossi-Hansberg, E., and Sarte, P.-D. (2018). The Impact of Regional and Sectoral Productivity Changes on the US Economy. *Review of Economic Studies*, 85(4), 2042–2096.

Diamond, R. and Suarez Serrato, J.C. (2025). Spatial Sorting and Inequality. NBER Working Paper No. 33609.

Donaldson, D. (2018). Railroads of the Raj: Estimating the Impact of Transportation Infrastructure. *American Economic Review*, 108(4–5), 899–934.

Eaton, J. and Kortum, S. (2002). Technology, Geography, and Trade. *Econometrica*, 70(5), 1741–1779.

Ellder, E. (2024). The 15-Minute City Dilemma? Balancing Local Accessibility and Gentrification in Gothenburg, Sweden. *Transportation Research Part D*, 135, 104360.

Ellder, E. (2025). Exploring Socio-Economic Inequalities in Access to the 15-Minute City Across 200 Swedish Built-Up Areas. *Journal of Transport Geography*, 122, 104057.

Ferrer-Ortiz, C., Marquet, O., Mojica, L., and Vich, G. (2022). Barcelona Under the 15-Minute City Lens: Mapping the Accessibility and Proximity Potential Based on Pedestrian Travel Times. *Smart Cities*, 5(1), 146–161.

Harris, C.D. (1954). The Market as a Factor in the Localization of Industry in the United States. *Annals of the Association of American Geographers*, 44(4), 315–348.

LeSage, J. and Pace, R.K. (2009). *Introduction to Spatial Econometrics*. CRC Press.

Moreno, C., Allam, Z., Chabaud, D., Gall, C., and Pratlong, F. (2021). Introducing the "15-Minute City": Sustainability, Resilience and Place Identity in Future Post-Pandemic Cities. *Smart Cities*, 4(1), 93–111.

Mouratidis, K. (2024). Time to Challenge the 15-Minute City: Seven Pitfalls for Sustainability, Equity, Livability, and Spatial Analysis. *Cities*, 153, 105274.

Padgham, M., Lovelace, R., Salmon, M., and Rudis, B. (2019). dodgr: An R Package for Network Flow Aggregation. *Findings*, Transport Findings. doi: 10.32866/6945.

Plaza-Herrera, D. and Mercade-Aloy, J. (2025). Inclusive 15-Minute Cities: An Age-Sensitive Assessment of Active Accessibility in the Metropolitan Area of Barcelona. *Cities*, 165, 106091.

Redding, S.J. (2016). Goods Trade, Factor Mobility and Welfare. *Journal of International Economics*, 101, 148–167.

Redding, S.J. and Rossi-Hansberg, E. (2017). Quantitative Spatial Economics. *Annual Review of Economics*, 9, 21–58.

Thaury, M.-O., Genet, S., Maurice, L., Tubaro, P., and Berkemer, S.J. (2024). City Composition and Accessibility Statistics in and Around Paris. *Frontiers in Big Data*, 7, 1354007.

UBE (2024). 15-Minute Cities: The Urban Planning Model Redefining How We Live. University of the Built Environment, London.